

BioSpecInfo

Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

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Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Purpose	Document the verification method, validation data for the spectral predictors, cross-device performance and declared model limitations.

1. Scope and V&V strategy

- **Verification:** the software is built correctly (syntax, no runtime errors, expected component behaviour).
- **Validation:** the scientific results (spectral predictors, descriptors) are compared against reference experimental databases.

2. Software verification (QA)

Technique	Description	Status
Syntax checking	<code>node --check</code> on the critical JavaScript blocks on every change.	✓ In use
End-to-end (E2E) testing	Automation with Playwright on headless Chromium: open sections, draw/load molecules, verify panels (notations, spectra, functional groups, ADMET, similarity), 2D/3D animations. Counting of console/page JavaScript errors (acceptance criterion: 0 non-network errors).	✓ In use
HTTP-server verification	RDKit WASM requires an HTTP context (CORS): tests run on a local server to reproduce the real environment.	✓ In use
Manual cross-device testing	Testing on real devices (see §4).	✓ In use
Automated unit-test suite (Vitest/Jest)	—	 Roadmap
Coverage ≥ 90%	—	 Roadmap

Every release is verified E2E with Playwright under a "zero JS errors" criterion before merging to the main branch.

3. Scientific validation of the spectral predictors

BioSpecInfo's spectral predictors use **heuristic group-additivity / structural rule** models (literature reference tables), chosen to guarantee **instant in-browser response** on any device.

3.1 ¹H-NMR benchmark vs SDBS

Metric	Reported value
Reference database	SDBS (Spectral Database for Organic Compounds, AIST)
Mean Absolute Error (MAE)	0.31 ppm
Signals within ± 0.5 ppm	89%

3.2 IR benchmark (C=O band) vs NIST

Metric	Reported value
Reference database	NIST Chemistry WebBook
Mean Absolute Error (MAE)	11 cm⁻¹
Bands within ± 30 cm ⁻¹	90%

3.3 Physicochemical properties

Descriptors (molecular weight, formula, LogP, TPSA, HBD/HBA, rings, stereocentres) are computed with **RDKit MinimalLib** and comparable with **PubChem** and **DrugBank**. Reproducible verification example: *cholesterol* $\rightarrow$ *C₂₇H₄₆O*, 4 rings, 8 stereocentres.

Benchmark data source: measurements carried out by the author as part of the undergraduate thesis. The values are to be understood as validation for **educational and qualitative functional-group identification** purposes, not as a substitute for reference quantum-mechanical calculations.

4. Cross-device performance

Smoothness tests of the interactive renderers (3D viewers, animations, editor) across a range of devices:

Device	Result
iPhone / iPad	30-60 fps
Windows / macOS laptop	60 fps
Raspberry Pi 4	30-60 fps

Size of the main `index.html` file: $\approx$ **4.5 MB** (the full package, including WASM modules and 3D assets, is larger but remains entirely static and cacheable offline).

5. Declared model limitations

In line with the principle of scientific transparency:

- Additivity model.** The NMR/IR predictors sum group contributions and do **not** fully model extended chemical-environment effects (long conjugation, anisotropic effects, intramolecular hydrogen bonds). Larger deviations are expected e.g. for **strongly conjugated carbonyls** or edge cases such as **formic acid**.
 - Qualitative/educational spectra.** The simulated spectra serve to recognise functional groups and understand structure-spectrum relationships, not to replace certified experimental data.
 - Network dependency for some functions.** IUPAC/common name, CAS number, GHS pictograms and real 3D models come from PubChem: without network the app shows a fallback and remains fully functional on local data.
 - Two-way accuracy.** For rigorous use, direct links to the official databases (NIST, SDBS, MassBank, nmrshiftdb2) are integrated into the interface.
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6. Release acceptance criteria

- ✓ `node --check` passes without errors on the modified modules.
 - ✓ Playwright E2E: 0 non-network JS errors.
 - ✓ No visible regression in the touched sections.
 - ✓ Service Worker cache version bumped.
 - ✓ `tools/verifica-farmaci.js`: no defects; every deviation recorded with its reason.
 - ✓ `verifica_guida`: whatever the documentation promises exists in the code.
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7. Verification battery — state at version `bsi-v178`

Verification is no longer only functional end-to-end testing: it includes benches dedicated to the properties a functional test does not observe (stability over time, network degradation, data accuracy).

7.1 Composition

Family	Benches	Subject
Scientific data	<code>verifica-farmaci</code> , <code>test_spettri</code> , <code>test_assi</code> , <code>test_assi_canvas</code> , <code>test_costanti</code> , <code>audit_dati</code>	structure vs molecular weight, functional-group recognition, axis conventions, physical constants

Family	Benches	Subject
AI agent	test_ko, test_404, test_503, test_firma, test_attesa, test_attesalunga, test_tetto, test_nucleo, browser_ko, browser_prova	retired model, CORS, overload, reasoning signatures, token ceiling, long waits, memory of unreachable providers
Stability	audit_stabilita, audit_promesse, audit_quota, test_filemanager, test_doppioinvio, test_sw, test_visore3d	long sessions, rejected promises, exhausted storage, unavailable archive, overlapping submissions, degraded network, WebGL contexts
Interface	browser_reset, browser_proxy, browser_proxyui, browser_rdkit, browser_lab, browser_frontiera, test_aggiorna, test_guidaproxy	panels, proxy, RDKit laboratory, updates
Consistency	verifica_guida	documentation ↔ code correspondence, conflict markers across 58 files

7.2 Properties verified that a functional test does not observe

Property	Method	Measured result
Absence of memory leaks	88 sections opened for 5 rounds, DOM node count per round	construction +26,876 nodes on the first round, +0 on the four that follow
Survival of exhausted storage	localStorage.setItem forced to throw	10 pages out of 10 stay operational
Network degradation (≠ no network)	requests held for 20 s, interception at context level	answer served from cache in 3,507 ms (threshold 3,500 ms)
Spectrum reproducibility	drawn twice, compared byte for byte	identical
WebGL contexts	viewer constructions counted across 10 molecules	from 7 down to 1

7.3 Notes on method

A bench that measures nothing passes, and that is the most insidious way to obtain a false guarantee. While the battery was being built, several cases of vacuous measurement were found and corrected, among them:

- Playwright's `page.route` does **not** intercept requests originated by the Service Worker: `context.route` is required. With the former, the "degraded network" test was measuring a perfectly healthy network and concluding in 16 ms.
- Probing `localStorage` exhaustion with a one-byte write proves nothing: after saturation, one byte always finds room.
- The ticks of an axis drawn on `<canvas>` are not readable as text: they must be collected by intercepting `fillText`, and the bottom-most line is the axis *title*, not a tick.

Hence the rule adopted: **every bench that simulates a condition must count how many times the simulation actually fired**, and fail if that count is zero.